Logarithmic and exponential forms of [redacted]ations

1 Rewrite each of the following equations in logarithmic form:

a $2^5 = 32$
b $6^x = 36$
c $9^1 = 9$
d $2^0 = 1$
e $(0.9)^0 = 1$
f $16^{1.5} = 64$
g $9^{\frac{3}{2}} = 27$
h $10^{-2} = 0.01$
i $3^{-4} = \dfrac{1}{81}$
j $x^{1.5} = 125$

2 Rewrite each of the following equations in exponential form:

a $\log_4 16 = 2$
b $\log_5 5 = 1$
c $\log_8 1 = 0$
d $\log_2 0.125 = -3$
e $\log_{5.8} 33.64 = 2$
f $\log_{\frac{1}{3}} 9 = -2$
g $\log_x 32 = 5$
h $\log_8 x = 6$
i $\log_3 \dfrac{1}{3} = -1$
j $\log_9 (3) = \dfrac{1}{2}$

3 True or false: The equation $4\left(1.09\right)^x = 20$ is equivalent to $x = \log_{1.09} 5$.

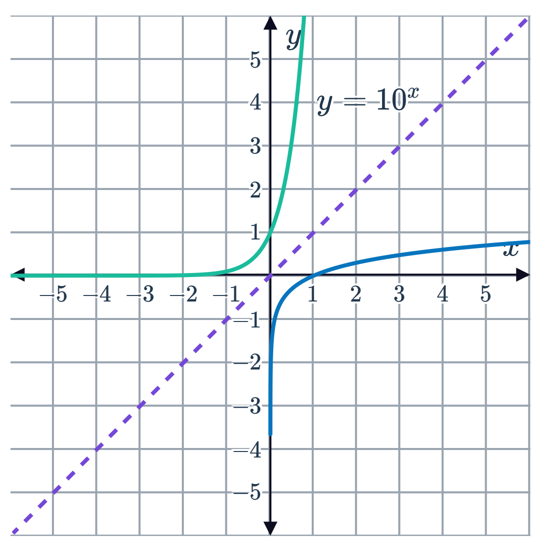
4 True or false: If $5^{x-6} = 2^{4x+3}$, then $x - 6 = \log_5 2^{4x+3}$.

Inverse relationship between exponentials and logarithms

5 Consider the graph of $y = 10^x$ and its reflection across the line $y = x$:

a Are the two curves inverse functions of each other?

b State the equation of the reflection of $y = 10^x$.



6 Consider the function $y = 2^x$.

a Sketch the graph of $y = 2^x$.

b On the same set of axes, sketch the graph of $y = \log_2 x$.

c Complete the following statement: The y-intercept of $y = 2^x$ is the same as the ▯ of $y = \log_2 x$.

7 Consider the function $y = 10^{-x}$:

a Sketch the graph of $y = 10^{-x}$.

b Find the equation of the inverse function of $y = 10^{-x}$.

c Sketch the graph of the inverse function.

8 Consider the function $f\left(x\right) = e^x - 3$ for all $x \in \mathbb{R}$.

a Find the inverse function y in terms of x.

b State the domain of the inverse as an inequality.

c State the range of the inverse using interval notation.

9 Consider the function $f\left(x\right) = \log_5\left(x - 3\right)$ for all $x > 3$.

a Find the inverse function y in terms of x.

b State the domain of the inverse function using interval notation.

c State the range of the inverse function using interval notation.

10 Consider the function $f\left(x\right) = 2e^x - 5$ for all $x \in R$.

a Find the inverse function y in terms of x.

b Using a graphing calculator, find the coordinates of the points at which the graphs of $y = f\left(x\right)$ and $y = f^{-1}\left(x\right)$ intersect. Round all values to three decimal places.